

SHARMIN AKTER, Ph.D.

Computational Physicist | Electronic Structure | Computational Quantum Science

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Professional Summary

Computational physicist and scientist with a Ph.D. in Computational Science and an M.S. in Physics, specializing in first-principles electronic-structure calculations, Density Functional Theory (DFT), self-interaction corrections, and computational modeling of molecular and materials systems. Experienced in numerical methods for quantum-mechanical simulations, scientific programming, high-performance computing, and electronic-structure software. Research includes molecular polarizabilities, ionization energies, core-level properties, and self-interaction-corrected DFT methods, with peer-reviewed publications in the *Journal of Chemical Physics*, *Physical Chemistry Chemical Physics*, *Physical Review A*, and *PNAS*. Experienced with C/C++, Python, Fortran, MPI, Unix/Linux, NRLMOL, and FLOSIC. Currently expanding this computational quantum-physics background into quantum algorithms and quantum simulation.

Research Experience

The University of Texas at El Paso, Department of Physics, El Paso, TX

2016-2023

Graduate Researcher / Computational Scientist

- Applied DFT and self-interaction-corrected methods to improve molecular and materials properties.
- Investigated self-interaction error and applied Perdew-Zunger SIC (PZSIC), Fermi-Löwdin orbital SIC (FLOSIC), and locally scaled SIC approaches.
- Computed and analyzed molecular polarizabilities, dipole moments, ionization energies, and core-level properties for atoms, ions, water clusters, polyacenes, and organic molecules.
- Worked with Fermi orbital descriptors, numerical optimization, computational linear algebra, and iterative electronic-structure calculations.
- Used electronic-structure workflows involving NRLMOL and FLOSIC, with scientific programming in C/C++, Fortran, Python, Unix shell scripting, and MPI-based parallel computing.
- Compared computational results with experimental and reference data to assess density-functional approximations and self-interaction corrections.
- Collaborated across computational-physics and electronic-structure projects and communicated results through peer-reviewed publications and conference presentations.

UTEP-PNNL Research Collaboration

2019-2023

Computational Electronic-Structure Research

- Investigated core ionization energies and electronic-structure properties of lanthanide chloride systems and benchmark organic molecules.
- Studied the effect of self-interaction error on core-level eigenvalues and ionization-energy predictions and compared calculations with experimental reference data.
- Applied first-principles electronic-structure methods to problems relevant to spectroscopy and materials characterization.

Technical Expertise

Quantum & Electronic Structure: DFT, self-interaction error/correction, PZSIC, FLOSIC, electronic-structure theory, molecular quantum mechanics, polarizability and response properties, ionization energies, core-level spectroscopy, computational materials science

Scientific Computing: Numerical analysis, numerical optimization, computational linear algebra, mathematical/computational modeling, scientific programming, high-performance computing, MPI, Unix/Linux

Programming: C/C++, Python, Fortran, MATLAB, R, Unix shell scripting, $\text{\LaTeX}$

Scientific Software & Tools: NRLMOL, FLOSIC, Git/GitHub, Jmol, Origin, scientific plotting and visualization; VASP – currently developing hands-on experience

Education

Ph.D. in Computational Science, The University of Texas at El Paso

2023

Dissertation: *Effect of Self-Interaction Correction on Molecular Polarizabilities and Core Ionization Energies*

Advisor: Tunna Baruah, Ph.D. | Co-advisor: Rajendra Zope, Ph.D.

M.S. in Physics, The University of Texas at El Paso

Thesis: *Self-Interaction Corrected Polarizabilities of Small Molecules*

M.Sc. in Physics, University of Dhaka

Thesis: *Study of Structural and Dielectric Properties of $\text{La}_{0.75}\text{Ca}_{0.05}\text{Sr}_{0.20-x}\text{Ba}_x\text{MnO}_3$ ($x=0.10$ and 0.15) Polycrystalline Perovskite*

B.Sc. in Physics, University of Dhaka

Selected Computational Coursework

Advanced Scientific Computing; Numerical Analysis; Numerical Optimization; Computational Methods of Linear Algebra; Mathematical and Computational Modeling; Advanced Algorithms; Machine Learning; Quantum Mechanics; Solid State Physics; Advanced Statistical Mechanics.

Selected Publications

1. S. Akter, J. Vargas, K. Sharkas, J. E. Peralta, K. A. Jackson, T. Baruah, and R. R. Zope, “How well do self-interaction corrections repair the over-estimation of molecular polarizabilities in density functional calculations?,” *Physical Chemistry Chemical Physics* (2021).
2. S. Akter, Y. Yamamoto, R. R. Zope, and T. Baruah, “Static dipole polarizabilities of polyacenes using self-interaction-corrected density functional approximations,” *The Journal of Chemical Physics* **154**, 114305 (2021).
3. S. Akter, Y. Yamamoto, C. M. Diaz, K. A. Jackson, R. R. Zope, and T. Baruah, “Study of self-interaction errors in density functional predictions of dipole polarizabilities and ionization energies of water clusters using Perdew–Zunger and locally scaled self-interaction corrected methods,” *The Journal of Chemical Physics* **153**, 164304 (2020).
4. K. Sharkas, K. Wagle, B. Santra, S. Akter, R. R. Zope, T. Baruah, K. A. Jackson, J. P. Perdew, and J. E. Peralta, “Self-interaction error overbinds water clusters but cancels in structural energy differences,” *PNAS* **117**, 11283–11288 (2020).
5. K. P. K. Withanage, S. Akter, C. Shahi, R. P. Joshi, C. Diaz, Y. Yamamoto, R. Zope, T. Baruah, J. P. Perdew, J. E. Peralta, and K. A. Jackson, “Self-interaction-free electric dipole polarizabilities for atoms and their ions using the Fermi–Löwdin self-interaction correction,” *Physical Review A* **100**, 012505 (2019).

Selected Research Presentations

- “Self Interaction Correction for Improved Description of Molecular Properties,” US–Africa Initiative Workshop in Electronic Structure, 2021.
- “Improving density functional calculations of molecular polarizabilities using locally scaled self-interaction corrections,” APS March Meeting, 2021.
- “Fermi–Löwdin orbital self-interaction corrections applied to water clusters: Polarizabilities, dipole moments, and ionization energies,” APS March Meeting, 2020.
- “Ionization potentials and static dipole polarizabilities of polyacenes using Fermi–Löwdin self-interaction corrected density functional approximation,” APS March Meeting, 2019.

Teaching Experience

The University of Texas at El Paso — Department of Physics

2016–2023

- Taught undergraduate and graduate courses including General Physics, Mechanics, Electromagnetism, Thermal Physics, Analytical Mechanics, Vibrations and Waves, Physical Optics, Astrophysics, and Computational Methods.
- Prepared lectures, assignments, quizzes, examinations, and instructional materials; supported laboratories and held office hours.

Honors & Professional Memberships

College of Science Research Assistantship, UTEP; Grace Hopper Celebration Scholarship/Travel Support; DOE-supported FOMMS Travel Support.

American Physical Society (APS); Society for Industrial and Applied Mathematics (SIAM); The FLOSIC Center.